MINITEN 25MG TABLET

Captopril, 25mg Tablet

Product description
White square shape crossed tablets.

Pharmacodynamic
MINITEN (Captopril) is an angiotensin converting enzyme (ACE) inhibitor with antihypertensive and vasodilating effects.

Pharmacokinetics
- Captopril is an orally active agent that does not require biotransformation for activity.
- The average minimal absorption is approximately 75%. The presence of food in the gastrointestinal tract reduces absorption by about 30-40%. Approximately 25-30% of the circulating drug is bound to plasma proteins.
- The apparent elimination half-life of unchanged Captopril in blood is about 2 hours. Greater than 95% of the absorbed dose is eliminated in the urine within 24 hours.
- Therefore, in patients with impaired renal function the dose should be reduced and/or dosage interval prolonged.

Indication
- Hypertension: Administered alone or combined with thiazide diuretics. The blood pressure lowering effects of captopril and thiazides are approximately additive.
- Congestive heart failure: In combination with a diuretic and when appropriate, digitalis and β-blockers.
- Myocardial infarction (MI) : In clinically stable patients with asymptomatic and symptomatic left ventricular dysfunction following MI.
- Diabetic nephropathy: In patients with type-I Insulin dependent diabetes mellitus. In these patients, captopril prevents the progression of renal disease and reduces associated clinical sequelae (dialysis, renal transplantation and death)

Recommended Dose
MINITEN should be taken one hour before meals. Dosage must be individualized.
Hypertension: Initiation of therapy requires consideration of recent antihypertensive drug treatment, the extent of blood pressure elevation, salt restriction, and other clinical circumstances. If possible, discontinue the patient's previous antihypertensive drug regimen for one week before starting **MINITEN**.
The initial dose of **MINITEN** is 50 mg once daily or 25 mg bid. If satisfactory reduction of blood pressure has not been achieved after one or two weeks, the dose may be increased to 100 mg daily in one or two divided doses. Concomitant sodium restriction may be beneficial when **MINITEN** is used alone.
If the blood pressure has not been satisfactorily controlled after one to two weeks at this dose, (and the patient is not already receiving a diuretic), a modest dose of a thiazide-type diuretic (e.g., hydrochlorothiazide, 25 mg daily), should be added. The diuretic dose may be increased at one-to two-week intervals until its highest usual antihypertensive dose is reached.
If **MINITEN** is being started in a patient already receiving a diuretic, **MINITEN** therapy should be initiated under close medical supervision, with dosage and titration of **MINITEN** as noted above.
If further blood pressure reduction is required, the dose may be increased incrementally (while continuing the diuretic) and a tid dosage schedule may be considered. The dose of **MINITEN** in hypertension usually does not exceed 150 mg/day. A maximum daily dose of 450 mg **MINITEN** should not be exceeded.
For patients with severe hypertension (e.g., accelerated or malignant hypertension), when temporary discontinuation of current antihypertensive therapy is not practical or desirable, or when prompt titration to more normotensive blood pressure levels is indicated, diuretic should be continued but other current antihypertensive medication stopped and **MINITEN** dosage promptly initiated at 25 mg bid or tid, under close medical supervision.
When necessitated by the patient's clinical condition, the daily dose of **MINITEN** may be increased every 24 hours or less under continuous medical supervision until a satisfactory blood pressure response is obtained or the maximum dose of **MINITEN** is reached. In this regimen, addition of a more potent diuretic e.g., furosemide, may also be indicated.
Beta-blockers may also be used in conjunction with **MINITEN** therapy, but the effects of the two drugs are less than additive.
Heart Failure: Initiation of therapy requires consideration of recent diuretic therapy and the possibility of severe salt/volume depletion. In patients with either normal or low blood pressure, who have been vigorously treated with diuretics and who may be hyponatremic and/or hypovolemic, a starting dose of 6.25 or 12.5 mg tid, may minimize the magnitude or duration of the hypotensive effect; for these patients, titration to the usual daily dosage can then occur within the next several days.
For most patients the usual initial daily dosage is 25 mg tid. After a dose of 50 mg tid is reached, further increases in dosage should be delayed, where possible, for at least two weeks to determine if a satisfactory response occurs. Most patients studied have had a satisfactory clinical improvement at 50 or 100 mg tid. A maximum daily dose of 450 mg of **MINITEN** should not be exceeded.
MINITEN should generally be used in conjunction with a diuretic and digitalis. **MINITEN** therapy must be initiated under very close medical supervision.

Myocardial Infarction: Therapy may be initiated as early as three days following a myocardial infarction. After an initial dose of 6.25 mg, captopril therapy should be increased to 37.5 mg daily in divided doses as tolerated. Captopril should then be increased as tolerated to 75 mg a day in divided doses during the next several days and to a final target dose of 150 mg daily in divided doses over the next several weeks.
If symptomatic hypotension occurs, a dosage reduction may be required. Subsequent attempts at achieving the target dose of 150 mg should be based on the patient's tolerance to captopril. Captopril may be used in patients treated with other post-, myocardial infarction therapies, e.g., thrombolytics, aspirin, beta blockers.

Diabetic Nephropathy: In patients with diabetic nephropathy, the recommended daily dose of captopril is 75 to 100 mg in divided doses.
If further blood pressure reduction is required, other antihypertensive agents such as diuretics, beta adrenoceptor blockers, centrally acting agents or vasodilators may be used in conjunction with captopril.

Dosage Adjustment In Renal Impairment:
Captopril in divided doses of 75 to 100 mg/day was well tolerated in patients with diabetic nephropathy and mild to moderate renal impairment. . Because **MINITEN** is excreted primarily by the kidneys, excretion rates are reduced in patients with impaired renal function. These patients will take longer to reach steady-state captopril levels and will reach higher steady-state levels for a given daily dose than patients with normal renal function. Therefore, these patients may respond to smaller or less frequent doses.
Accordingly, for patients with significant renal impairment, initial daily dosage of **MINITEN** should be reduced, and smaller increments utilized for titration, which should be quite slow (one-to two-week intervals).
After the desired therapeutic effect has been achieved, the dose should be slowly back-titrated to determine the minimal effective dose. When concomitant diuretic therapy is required, a loop diuretic (e.g., furosemide), rather than a thiazide diuretic, is preferred in patients with severe renal impairment.

Mode of action
MINITEN (Captopril) is an angiotensin converting enzyme (ACE) inhibitor with antihypertensive and vasodilating effects.

Contraindication
- History of hypersensitivity to Captopril, or any other ACE inhibitor.
- History of angioedema associated with previous ACE inhibitor therapy.
- Hereditary / idiopathic angioneurotic oedema.

Precautions and Warning
Precaution :-
General: Impaired Renal Function:
Some patients with renal diseases particularly those with bilateral renal artery stenosis, have developed increase in BUN and serum creatinine after reduction of blood pressure with Captopril, usually along with a diuretic. Captopril dosage reduction or discontinuation of diuretic, or both maybe required. For some of these patients, it may not be possible to normalize blood pressure and maintain adequate renal perfusion. Some patients with heart failure have experienced reduction in renal function during long-term treatment that usually stabilized at the reduced level. A few patients have developed angioedema of the face, mucous membrane of the mouth and the extremities which is reversible upon discontinuation of the drug. Laryngeal edema has also been reported.

Valvular stenosis: Captopril, as with any drug that reduces vascular resistance, should be used only with extreme caution in patients with aortic stenosis because of the potentially harmful consequence of reduced coronary perfusion secondary to reduced blood pressure.

Hyperkalemia: Elevations in serum potassium have been observed in some patients treated with ACE inhibitors including Captopril. When treated with ACE inhibitors, patients at risk of development of hyperkalemia include those with: Renal insufficiency; diabetes mellitus; and those using concomitant potassium-sparing diuretics, Potassium supplements or potassium containing salt substitutes; or other drugs associated with increases in serum potassium (eg: heparin) Cough: Cough has been reported with the use of ACE inhibitors. The cough is nonproductive, persistent and resolves after discontinuation of therapy.

Surgery/Anesthesia: In patients undergoing major surgery or during anesthesia with agents that produce hypotension, Captopril will block angiotensin II formation secondary to compensatory renin release. If hypotension occurs and is considered to be due to this mechanism, it can be corrected by volume expansion.

Patients should be advised to their physician any signs and symptoms suggesting angioedema (eg, swelling of face, eyes. lips. tongue, larynx and extremities, difficulty in swallowing or breathing, hoarseness), and to discontinue therapy. Patient should be told to report promptly any indication of infection (eg, sore throat, fever) that does not respond promptly to standard therapy which may be sign of neutropenia, or of progressive edema which might be related to proteinuria and nephrotic syndrome.

All patients should be cautioned that excessive perspiration and dehydration may lead to excessive fall in blood pressure because of reduction in fluid volume. Other causes of volume depletion: vomiting or diarrhea, may also lead to fall in blood pressure; patients should be advised to consult with physician.

Patients should be warned against interruption or discontinuation of medication unless instructed by the physician.

Patients should be informed that Captopril should be taken 1hr before meals.

Captopril may cause a false positive urine test for acetone.

Jan 2018
M3-14-1647

Use in Pregnancy:
Feta/Neonatal Morbidity and Mortality: When used in pregnancy during the 2nd and 3rd trimesters, ACE inhibitors can cause injury and even death to the developing fetus. When pregnancy is detected, Captopril should be discontinued as soon as possible.

Use In lactation:
Small amounts of Captopril are excreted in human breast milk. Therefore breastfeeding should be avoided.
Use in Children: Safety and effectiveness in children have not been established.

WARNINGS:
ACE inhibitors have been shown to be strongly fetotoxic in animal studies.
Recently available data indicate that ACE inhibitors can cause foetal and neonatal morbidity and mortality when administered to pregnant woman.
The use of these agents during pregnancy is not recommended.

INCREASED RISK OF BIRTH DEFECTS, FETAL AND NEONATAL MORBIDITY AND DEATH WHEN USED THROUGHOUT PREGNANCY.

Proteinuria has been seen < 1% of patient receiving Captopril, but this has been predominantly in patients had prior renal diseases. Although membranous glomerulopathy in biopsies taken from some proteinuric patients, or received relatively high doses of Captopril (in excess of 150mg/day). or both. For patients with prior renal diseases or those receiving Captopril at doses > 150 mg/day should have urinary protein estimations (dipstick on first morning urine) prior to treatment and periodically thereafter.

Neutropenia has occurred in some patients in the clinical studies within 3 months after Captopril has been started, especially for those who has preexisting impaired renal function collagen vascular disease, or who are receiving immunosuppressant therapy, the while blood cell and differential counts should be performed prior to therapy, every 2 weeks during the first 3 months of Captopril therapy and periodically thereafter.

If the neutrophil count falls below 1000/mm3.Captopril should be discontinued and the patient course should be followed. All patients receiving Captopril should be told to report any sign of infection (eg: sore throat. fever). Serious infections resulting from neutropenia and which proved fatal in a few occurred only in patients with impaired renal function.

Hypotension: Excessive hypotension was rarely seen in hypertensive patients but is a possible consequence of Captopril use in salt-/volume depleted patients (eg. those treated vigorously with diuretics), patients with heart failure or those patients undergoing dialysis. In hypertension, the possibility of hypotensive effects with the initial doses of Captopril can be minimized by either discontinuing the diuretic or increasing the sail intake approximately 1 week prior to initiation of treatment with Captopril, or initiating of therapy with small doses (6.25 or 12.5mg). Medical supervision should be provided for at least 1 hr after the initial dose.

In patients with heart failure exaggerated hypotensive have been occurred usually within 1 hr of the initial dose of Captopril. This transient fall in pressure may occur after any of the first several doses and is usually well tolerated, producing either no symptoms or brief mild lightheadedness, although in rare instances it has been associated with arrhythmia or conduction defects.

Because of the potential fail in blood pressure in these patients, therapy should be started under very close medical supervision. A starting dose 6.25 or 12.5mg twice or 3 times daily may minimize the hypotensive effect.

Hepatic failure: Rarely ACE inhibitors have been associated with a syndrome that starts with cholestalic jaundice and progresses to fulminant hepatic necrosis and (sometimes) death. The mechanism of this syndrome is not understood. Patients receiving ACE inhibitors who develop jaundice or markedly elevations or liver enzymes should discontinue the ACE inhibitor and receive appropriate medical follow-up.

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Interaction with other medicaments
- Alcohol or general anaesthetics may produce additive hypotensive effect.
- Concomitant use of other commonly used anti-hypertensive agents (e.g: β-blockers and long-acting calcium channel blockers) may increase the hypotensive effects of Captopril.
- Treatment with nitroglycerine and other nitrates, or other or other vasodilator, should be used with caution.
- Diuretics (thiazide or loop diuretic) : Risk of volume depletion and of hypotension when initiating therapy with Captopril. No clinically significant drug interaction have been found.
- Phenothiazines increase the hypotensive effects of Captopril.
- NSAIDS including Indomethacin exert an additive effect on the increase in serum potassium whereas renal function may decrease. Acute renal failure may occur rarely, particularly in patients with compromised renal function such as the elderly or dehydrated. Chronic administration of NSAIDS may reduce the antihypertensive effect of an ACE inhibitor.
- Antacids may reduce Captopril antihypertensive effects.
- Sympathomimetics : may reduce the antihypertensive effects of ACE inhibitors; patients should be carefully monitored.
- Potassium-sparing diuretics (e.g: Spironolactone, Triamterene or Amiloride).
- Potassium-containing medications may result in hyperkalaemia.
- Digoxin and Lithium: Increased risk of reversible toxicity caused by ACE Inhibitors.
- Captopril may cause a false-positive test result for urine acetone.
- ACE Inhibitor, including captopril, can potentiate the blood glucose-reducing effects of insulin and oral antidiabetics (such as Sulphonyiurea) in diabetics.
- ACE Inhibitor may enhance the hypotensive effects of certain tricyclic antidepressants and antipsychotics. Postural hypotension may occur.
- Allopurinol, Procainamide, Cytostatic or immunosuppressive agent: concomitant administration with ACE Inhibitors may lead to an increased risk for leucopenia especially when the later are used at higher than currently recommended doses.

Pregnancy and Lactation.
- Pregnancy : Captopril is of category C during the first trimester, and ofcategory D during second and third trimester, thus it can cause injury and even death to the fetus. When pregnancy is detected, Captopril should be discontinued.
- Lactation : Captopril concentration in breast milk is about 1% of those in maternal blood. Therefore breast feeding should be avoided.

INCREASED RISK OF BIRTH DEFECTS, FETAL AND NEONATAL MORBIDITY AND DEATH WHEN USED THROUGHOUT PREGNANCY

Side effects
- Hypotensive at the initiation of therapy, hyperkalaemia, hypoglycaemis, skin rash, alopecia, itching, angioedema, urticaria, Steven Johnson syndrome, erythema multiforme. Photosensitivity, erythroderma, exfoliative dermatitis.
- Sleep disorders, confusion, depression, taste impairment, dizziness, headache, tiredness, paraesthesia, vertigo, malaise, fever, chest pain, nausea, vomiting, gastric, irritations, anorexia, gastrointestinal pain, diarrhoea, constipation, dry mouth, stomatitis, glossitis, peptic ulcer, pancreatitis.
- Impared hepatic function and cholestasis (including jaundice), hepatitis including necrosis, elevated hepatic enzymes and bilirubin.
- Chronic persistent dry cough, dyspnoea, bronchospasm, rhinitis, allergic alveolitis / eosinophilic pneumonia.
- Neutropenia / agranulocytosis, pancytopenia particularly in patients with renal dysfunction, anaemia (including aplastic and haemolytic), thrombocytopenia, lymphadenopathy, eosinophilia, stroke. Syncope, blurred vision, tachycardia or tachyarrhythmia, angina pectoris, palpitations, cardiac arrest, cardiogenic shock, Raynaud's syndrome, flush, pallor, myalgia, arthralgia, renal failure, polyuria, oliguria, nephritic syndrome, impotence, gynaecomastia.
- Investigations: very rare: proteinuria, eosinophilia, increased serum potassium, decreased serum sodium, elevation of BUN, serum creatinine and serum bilirubun, decreases in haemoglobin, haematocrit, leucocytes, thrombocytes.
- Stop taking this drug in the following cases and immediately contact your doctor: Swelling of the face, lips, tounge and/or larynx which entail difficulties in breathing or swallowing.

Symptoms and treatment overdose
- Symptoms of overdosage are: Severe hypotension, shock, stupor, bradycardia, electrolyte disturbances and renal failure.

If major hypotension is noted, the patient should lie down with raised legs.
- Measures to prevent absorption can be done : e.g. Gastric lavage, administration of adsorbent and sodium sulphate within 30 minutes after Captopril intake.
- Captopril is removable by hemodialysis. The hypotensive effect is best corrected by volume expansion with IV infusion of normal saline. Bradycardia or extensive vagal reactions should be treated by administering Atropine.

Storage condition
Store up to 30 °C

Shelf life
3 years

Pack size
20, 100 and 1000 Tablets

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This is a medicament
- Medicament is a product which affects your health, and its consumption contrary to instructions is dangerous for you.
- Follow strictly the doctor's prescription, the method of use and the instructions of the pharmacist who sold the medicament.
- The doctor and the pharmacist are experts in medicine, its benefits and risks.
- Do not by yourself interrupt the period of treatment prescribed for you.
- Do not repeat the same prescription without consulting your doctor.
- Keep medicament out of the reach of children.

COUNCIL OF ARAB HEALTH MINISTERS
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